

Government, Taxation and Economic Complexity[†]

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Abstract

This paper investigates the relationship between tax complexity, economic complexity, and government size. The findings are obtained through cross-country panel data and two-way fixed effects regressions, and reveal a nuanced connection: under small (large) government scales, increased tax complexity is associated with greater (smaller) economic sophistication. This suggests that tax complexity's positive and negative aspects vary depending on the government conditions provided to the economy.

Keywords: Economic Complexity; Tax Complexity; Government Size; Taxation.

JEL Classifications: H21; O39; M48.

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1 Introduction

A substantial debate surrounds the effect of taxation on economic growth ([Vermeend et al., 2008](#)). An important component of a tax system is its degree of intricacy, difficulty, and convolution, known as tax complexity. On one hand, more complex tax structures can be a necessary condition for modern economies to maintain economic growth ([Easterly and Rebelo, 1993](#)). Conversely, tax complexity may be negative due to an overload of information, confusion, and uncertainty ([Thomas Hoppe and Sureth-Sloane, 2023](#)). There are challenges to improving tax systems, such as undergoing administrative protocols, political conflicts, and the potential of rent-seeking ([Congleton and Hillman, 2015](#)).

A novel approach to contemplating economic growth involves evaluating the variety and sophistication of goods produced in the economy, a concept referred to as economic complexity ([Hidalgo et al., 2007](#)). In this paper, we investigate the effect of tax complexity on economic complexity, taking into account the role of government size as a mediator in this relationship.

We use cross-country panel data and two-way fixed effects regressions to find that the effect of tax complexity on economic sophistication crucially depends on government size. For small (high) government scales, a higher tax complexity is significantly associated with greater (smaller) economic sophistication. We interpret this finding as evidence that the positive and negative aspects of tax complexity can manifest in distinct ways based on the conditions provided by the government to the economic environment.

This paper is situated in the intersection between complexity economics (e.g., [Hi-](#)

dalgo et al., 2007; Hausmann et al., 2007) and taxation (e.g., Easterly and Rebelo, 1993; Vermeend et al., 2008) . Lapatinas et al. (2019) is a close reference, studying how economic complexity in OECD countries depend on the emphasis of their tax systems on labor or capital. Our paper contributes to the literature on the relationship between tax and economic complexity by considering the moderator role of government size and incorporating non-OECD countries.

2 Data and Method

Tax complexity is measured by the Tax Complexity Index (*TaxComplex*), which is based on the complexity of a country's corporate income tax system, particularly as faced by multinational corporations. It is available for the years 2016, 2018, and 2020. Economic complexity is represented by the Economic Complexity Index (*ECI*). Finally, government size is measured by government spending and tax revenue. For a detailed description of these indicators and other control variables listed below, including sources, please refer to the Appendix.

The country-level regression model is

$$ECI_{it} = \alpha + \beta_1(Z_{it} \times TaxComplex_{it}) + \beta_2 TaxComplex_{it} + \beta_3 Z_{it} + \mathbf{X}'_{it}\theta + \phi_i + \tau_t + \varepsilon_{it}, \quad (1)$$

where the subscripts i and t refer, respectively, to country and year; Z_{it} is government spending or tax revenue; ϕ_i and τ_t are, respectively, individual (country) fixed effects and time fixed effects; $\mathbf{X}_{it}$ is a vector of control variables composed of population density (inhabitants per km² in logs), per-capita income (constant 2015 US\$, in logs), Rule of Law

index, internet coverage as a percentage of the population, average years of schooling, work productivity, share of bank assets, financial access, and bank returns. These variables were selected based on previous research recommendations (Nguyen et al., 2020; Barros Jr et al., 2022; Bahrami et al., 2023).

We are interested on the average impact of *TaxComplex* on *ECI*, which is given by $\beta_1 Z_{it} + \beta_2$. As a result, we are particularly interested in the following hypothesis tests: $\mathcal{H}_0 : [\beta_1 \ \beta_2] = \mathbf{0}$ (no relationship) and $\mathcal{H}_0 : \beta_1 = 0$ (impact, if any, does not depend on government size measures).

3 Results

Table 1 displays the regression estimates. The eight columns, divided between panels A and B, correspond to different government size measures and sets of control variables.

Regarding the joint test for β_1 and β_2 , we reject the null hypothesis in half of the cases, at the 5% level. Furthermore, in most cases, the individual tests reject $\beta_i = 0$, for $i = 1, 2$, at the 5% level. Thus, there is evidence that tax complexity is related to ECI and that the government size mediates such relation.

Given that β_1 and β_2 have opposite signs, it is especially important to estimate the marginal effect of tax complexity on ECI, which depends on the government size measure. Considering the range from the minimum to the maximum value of $Z_{i,t}$, we evaluate the marginal effects of tax complexity on ECI using the most general specifications reported in Table 1. The marginal effects are displayed in Table 2. In all cases, there is evidence that for low values of the government size proxies, the marginal effect of tax complexity on ECI is

positive and significant at the 5% level. However, for high values of the government size proxies, the sign of the marginal effect becomes negative and significant at the 5% level in most specifications. Given that the estimates for β_1 are negative, this is not surprising. Therefore, there is evidence suggesting that tax complexity is negatively related to ECI when government size is large.

The findings support the notion that when government size is relatively small, improvements in tax complexity result in effective economic sophistication gains. Conversely, when governments are too large, changes in tax structure may be related to ambiguous rules, numerous exceptions, or even resources wasted in administrative processes or political negotiations, resulting in adverse effects on economic complexity.

4 Conclusion

This paper explores the connection between tax complexity and economic sophistication, emphasizing the moderating role of the government. The contribution lies in showing that, while governments with limited involvement in the economy enhance the positive impact of tax complexity on economic sophistication, the reverse relationship can be observed in environments where governments are more prevalent. Future research can further investigate the mechanics behind those findings.

References

Bahrami, F., B. Shahmoradi, J. Noori, E. Turkina, and H. Bahrami (2023). Economic complexity and the dynamics of regional competitiveness a systematic review. *Competitive-*

ness Review 33(4), 711–744.

Barros Jr, F., L. Brotherhood, and V. Rodrigues (2022). Economic complexity and corporate governance. *Applied Economics Letters* 29(1), 68–71.

Congleton, R. D. and A. L. Hillman (2015). *Companion to the Political Economy of Rent Seeking*. Edward Elgar Publishing.

Easterly, W. and S. Rebelo (1993). Fiscal policy and economic growth. *Journal of Monetary Economics* 32(3), 417–458.

Hausmann, R., J. Hwang, and D. Rodrik (2007). What you export matters. *Journal of economic growth* 12, 1–25.

Hidalgo, C. A., B. Klinger, A.-L. Barabási, and R. Hausmann (2007). The product space conditions the development of nations. *Science* 317(5837), 482–487.

Lapatinas, A., A. Kyriakou, and A. Garas (2019). Taxation and economic sophistication: Evidence from OECD countries. *PLOS ONE* 14(3), e0213498.

Nguyen, C. P., C. Schinckus, and T. D. Su (2020). The drivers of economic complexity: International evidence from financial development and patents. *International Economics* 164, 140–150.

Thomas Hoppe, Deborah Schanz, S. S. and C. Sureth-Sloane (2023). The tax complexity index - a survey-based country measure of tax code and framework complexity. *European Accounting Review* 32(2), 239–273.

Vermeend, W., R. van der Ploeg, and J. W. Timmer (2008, April). *Taxes and the economy*.
Cheltenham, England: Edward Elgar Publishing.

5 Appendix

Table 1: Regression Results

Panel A: Tax Complexity and Tax Revenue				
	(1)	(2)	(3)	(4)
Tax Complexity x Tax Revenue	-0.088* (0.048)	-0.13*** (0.037)	-0.12*** (0.041)	-0.13*** (0.040)
Tax Revenue	0.030 (0.028)	0.049** (0.020)	0.036 (0.022)	0.033 (0.020)
Tax Complexity	1.88* (0.99)	2.74*** (0.85)	2.81*** (0.82)	2.76*** (0.90)
Control	Year Controls	Year x Controls	Year Controls	Year x Controls
Work productivity	No	No	Yes	Yes
N	166	166	131	131
R^2 within	15%	42%	22%	47%
# Countries	72	72	51	51
p -Value of $\mathcal{H}_0 : [\beta_1 \beta_2] = \mathbf{0}$	0.170	0.003	0.004	0.010
Panel B: Tax Complexity and Government Spending				
	(5)	(6)	(7)	(8)
Tax Complexity x Gov. Spending	-0.060 (0.042)	-0.093** (0.046)	-0.094** (0.047)	-0.14** (0.056)
Gov. Spending	0.081*** (0.020)	0.083*** (0.024)	0.095*** (0.023)	0.10*** (0.028)
Tax Complexity	1.10 (0.78)	1.78** (0.85)	1.87** (0.88)	2.68** (1.04)
Control	Year Controls	Year x Controls	Year Controls	Year x Controls
Work productivity	No	No	Yes	Yes
N	177	177	137	137
R^2 within	30%	42%	31%	48%
# Countries	79	79	53	53
p -Value of $\mathcal{H}_0 : [\beta_1 \beta_2] = \mathbf{0}$	0.364	0.115	0.113	0.044

Notes: ***, **, * mean significant at 1%, 5% and 10% level, respectively. Clustered robust std. error in parenthesis. "Year" and "Control" refer to time dummies and the addition of control variables, respectively. "Year x Control" indicates the interaction between the control variables and the time dummies.

Table 2: Marginal Effect of Tax Complexity on ECI

Specification	(2)		(4)		Specification	(6)		(8)	
Tax Revenue at	ME	S.E.	ME	S.E.	Gov. Spending at	ME	S.E.	ME	S.E.
7	1.83***	0.61	1.96***	0.76	5.3	1.29**	0.62	1.89***	0.64
9.5	1.51***	0.53	1.65**	0.65	7.6	1.08**	0.53	1.58***	0.56
12	1.18***	0.46	1.34**	0.54	9.9	0.86*	0.44	1.26***	0.47
14.5	0.86**	0.39	1.02**	0.43	12.2	0.65*	0.37	0.95**	0.4
17	0.53	0.34	0.71**	0.35	14.5	0.43	0.31	0.64*	0.34
19.5	0.21	0.3	0.4	0.3	16.8	0.22	0.28	0.33	0.3
22	-0.12	0.3	0.09	0.3	19.1	0.1	0.3	0.01	0.28
24.5	-0.44	0.32	-0.23	0.35	21.4	-0.21	0.34	-0.3	0.31
27	-0.76**	0.36	-0.54	0.44	23.7	-0.42	0.41	-0.61*	0.36
29.5	-1.09***	0.42	-0.85	0.54	26	-0.64	0.5	-0.92**	0.42
32	-1.41***	0.5	-1.16*	0.65	28.3	-0.85	0.59	-1.24**	0.5

Notes: ME refers to marginal effect and S.E. to standard errors. ***, **, * mean significant at 1%, 5% and 10% level, respectively. The values for “Tax Revenue” and “Gov. Spending” are based on deciles: the first value is the minimum, the second value is the first decile, the third value is the second decile, etc.

Variable	Description
ECI	Economic Complexity Index, or ECI, is a measure of an economy's capacity which can be inferred from data connecting locations to the activities that are present in them. Font: Observatory of Economic Complexity
Tax Complexity Index	The Tax Complexity Index measures the complexity of a country's corporate income tax system as faced by multinational corporations. It covers the complexity of the tax code and the complexity of the tax framework. Font: taxcomplexity.org
Tax Revenue	Tax revenue (% of GDP). Tax revenue refers to compulsory transfers to the central government for public purposes. Font: World Bank
Government Spending	General government final consumption expenditure (% of GDP). Source: World Bank
Internet	Individuals using the Internet (% of population). Source: World Bank
Per capita GDP	GDP per capita (constant 2015 US\$). Used as natural logarithm. Source: World Bank.
Population Density	Population density (people per sq. km of land area). Source: World Bank
Education	Average Years of Schooling. Source: Our World in Data
Rule of Law	Reflects perceptions of the extent to which agents have confidence in and abide by the rules of society, and in particular the quality of contract enforcement, property rights, the police, and the courts, as well as the likelihood of crime and violence. Source: World Governance Indicators
Productivity	Productivity is measured as GDP per hour of work. Source: Penn World Table
Bank Asset Share	Total assets held by deposit money banks as a share of the sum of deposit money bank and Central Bank claims on the domestic non financial real sector. Source: World Bank Global Financial Development Database
Financial Access	Automated teller machines (ATM's) per 100,000 adults. Used as natural logarithm. Source: World Bank Global Financial Development Database.
Bank Returns	Commercial banks' after-tax net income to yearly averaged total assets. Source: World Bank Global Financial Development Database